



# Prioritising Institutional and Technological Interventions for Engineering Project Credit-Risk Governance: An Integrated DEMATEL–ISM–MICMAC and Exploratory System Dynamics Approach

Shibin Zhong<sup>1</sup>, Zaohong Zhou<sup>1\*</sup>, Yanqing Huang<sup>2</sup>, Wenshu Zhou<sup>1</sup>

<sup>1</sup> School of Information Management and Mathematics, Jiangxi University of Finance and Economics, 330013 Nanchang, China

<sup>2</sup> Digital Jiangxi Technology Co., Ltd., 330038 Nanchang, China

\* Correspondence: Zaohong Zhou (Zhouzh5652@163.com)

Received: 07-16-2026

Revised: 09-04-2026

Accepted: 09-14-2026

**Citation:** S. B. Zhong, Z. H. Zhou, Y. Q. Huang, and W. S. Zhou, “Prioritising institutional and technological interventions for engineering project credit-risk governance: An integrated DEMATEL–ISM–MICMAC and exploratory system dynamics approach,” *J. Oper. Strateg. Anal.*, vol. 4, no. 3, pp. 196–219, 2026. <https://doi.org/10.56578/josa040305>.



© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

**Abstract:** Engineering project credit-risk governance requires regulators and project participants to coordinate institutional controls with digital supervision capabilities. Yet decision-makers often lack a structured basis for identifying the factors that should receive priority and for judging how institutional and technological interventions may perform over time. This study investigates the causal structure of engineering project credit risk and examines the policy implications of alternative governance interventions. An online questionnaire collected 86 complete responses covering 33 predefined directional relationships among 13 factors organised under the Technology–Organization–Environment (TOE) framework. Full-precision mean scores were analysed using the Decision-Making Trial and Evaluation Laboratory (DEMATEL), Interpretive Structural Modeling (ISM), and Matrix of Cross-Impact Multiplications Applied to Classification (MICMAC). An exploratory system dynamics (SD) model was then used to compare the baseline, institutional-response, technology, and combined scenarios. Robustness was examined through alternative response coding, threshold sensitivity tests, and 1,000 bootstrap resamples. The results showed that insufficient credit verification by supervision units, environmental and resource compliance risk, and lagging credit-management methods were the three most prominent factors. The ISM analysis placed environmental and resource compliance risk at the root of the four-level hierarchy, while MICMAC classified four factors as independent drivers. All bootstrap samples retained the same three leading factors, and alternative coding preserved the complete prominence ranking (Spearman’s  $\rho = 1.000$ ). In the exploratory simulation, the technology intervention produced a credit index of 39.09 at time 20, compared with 4.06 under the baseline scenario. The institutional-response intervention showed no clear long-horizon advantage. At time 50, the combined scenario produced a value of 33.73, only slightly higher than the technology-only value of 33.42. These findings indicate that engineering project credit-risk governance should prioritise verifiable supervision, interoperable monitoring, and timely credit-management processes. The integrated framework provides a transparent basis for intervention prioritisation and lifecycle governance, while the simulation results should be interpreted as policy experiments rather than industry forecasts.

**Keywords:** Engineering project credit risk; Governance intervention prioritisation; Smart supervision; Causal decision analysis; DEMATEL–ISM–MICMAC; Exploratory system dynamics

## 1 Introduction

Engineering project credit risk arises when project participants fail to meet contractual, regulatory, quality, safety, schedule, cost, or environmental obligations. Institutions establish the formal and informal rules that shape such behaviour [1]. In construction management, institutional theory has been used to examine governance arrangements [2], environmentally responsible adoption [3], responsive multi-party regulation [4], and the interaction between regulation and digital technology [5]. Digital artefacts can carry or initiate institutional change [6], but

the digital transformation of construction still depends on the alignment of technological capability, organisational readiness, and external support [7]. For regulators and project organisations, the central decision problem is therefore not whether digital technologies should be adopted in isolation, but how institutional arrangements and smart supervision technologies should be coordinated and prioritised to control credit risk throughout the project lifecycle.

Institutional theory provides a meta-theoretical basis for understanding the regulatory setting in which these decisions are made. North [1] defines institutions as the “rules of the game” in society, covering both formal institutions, such as laws and regulations, and informal institutions, such as industry norms and credit culture. Qiu and Chen [2] systematically reviewed the application of institutional theory in construction project management and distinguished its regulative, normative, and cognitive dimensions. Ayres and Braithwaite [4] proposed a tripartite regulatory framework involving government, enterprises, and public-interest groups, arguing that third-party participation can strengthen regulatory efficacy. However, institutional rules do not automatically produce effective supervision. They must be translated into operational responsibilities, verifiable procedures, information flows, and enforceable responses before they can function as practical regulatory measures [6]. This translation creates a resource-allocation problem because decision-makers must determine which institutional weaknesses require attention and which technological capabilities are most likely to support their implementation.

Recent engineering research demonstrates why institutional and technological measures need to be considered jointly. Studies of blockchain and smart contracts have addressed payment reliability and the reconfiguration of trust [8, 9]; digital twins and connected sensing have been used for real-time condition monitoring [10, 11]; and reviews of artificial intelligence have documented both its potential and its implementation limits in architecture, engineering, and construction [12, 13]. Research on contract claims has also linked delays, cost overruns, and quality failures to performance risk [14], while research on digitally enabled construction management has shown that governance and organisational coordination remain consequential in technologically advanced project environments [15]. These studies identify relevant technological and contractual mechanisms, but they do not establish which institutional, organisational, and technological constraints should be addressed first. Nor do they show whether an intervention that appears influential in a static causal structure will retain its relative value when feedback effects and implementation delays are considered.

The Technology–Organization–Environment (TOE) framework provides a conceptual structure for organising the conditions that influence technology adoption [16]. Related construction research has examined payment automation [17], the causes of schedule and cost overruns [18], dynamic modelling of site risks [19], safety prediction [20], governmental and industry applications of blockchain [21], and the mediating role of relational governance in digital project resilience [22]. Taken together, this literature indicates that smart supervision cannot operate as a stand-alone technical system. Its effectiveness depends on clearly assigned participant responsibilities, interoperable data, organisational capacity, and enforceable institutional rules. In this study, the TOE framework is used to organise the factor set before analysis. It does not determine the causal hierarchy, the relative importance of the factors, or the interventions that should receive priority; these are treated as empirical decision questions.

Smart supervision research examines how emerging technologies alter regulatory processes, information exchange, and the allocation of supervisory responsibilities. Wu et al. [21] reviewed the mechanisms through which blockchain can support efficiency and sustainability in construction. Liu and Wang [22] found that relational governance mediates the relationship between emerging digital technologies and construction-project resilience. Despite this progress, an imbalance remains between technical feasibility and the institutional and organisational work needed to make supervisory information usable in practice. A monitoring platform may collect extensive project data, for example, but those data have limited regulatory value when verification duties are unclear, systems are not interoperable, or identified defaults do not trigger a defined response. The management question is therefore how institutional and technological interventions can be combined within an auditable governance process rather than how a particular technology performs on its own.

Methodologically, integrated decision models have begun to connect structural judgements with dynamic behaviour. Decision-Making Trial and Evaluation Laboratory–System Dynamics (DEMATEL–SD) models have been used to examine project-cost factors [23] and schedule-delay controls [24]; Decision-Making Trial and Evaluation Laboratory–Interpretive Structural Modeling–System Dynamics (DEMATEL–ISM–SD) has been applied to smart-city carbon emissions [25]; and fuzzy DEMATEL–SD has been used to study social sustainability in infrastructure projects [26]. The unresolved methodological issue in engineering project credit-risk governance is not the simple combination of several established techniques. It is the construction of a traceable analytical chain from a clearly bounded set of questionnaire judgements to causal prominence, hierarchical position, driving dependence, and dynamic intervention behaviour. Such a chain is necessary if the results are to support defensible priority setting. It must also disclose which relationships were assessed, how the structural threshold was selected, which factors entered the dynamic model, and which factors were omitted.

Evidence for the factor set was drawn from research on collaborative environmental, social, and governance

practices [27], barriers to construction robotics and digital twins [28, 29], governmental supervision technologies [30], construction risk analytics [31–33], and contractual risk allocation [34]. Additional evidence came from studies of supply-chain collaboration [35], barriers to technology adoption [36, 37], digital tendering [38], studies of China’s green-construction transition and construction-industry digital-transformation policies [39, 40], and data-governance requirements in sustainable smart cities [41]. These sources support the inclusion of institutional, organisational, technological, contractual, environmental, and project-execution factors within a common decision structure. The available study records document the iterative synthesis and consolidation of these factors and the construction of the final 13-factor instrument. They do not, however, contain a defensible count of the initial candidate pool or a formal record of factor deletion and expert consultation. The account of factor development is therefore restricted to procedures that can be verified from the available records rather than reconstructed retrospectively.

Against this background, the study addresses three questions: (1) Which factors have the greatest prominence and causal influence within the traceable direct-relation network? (2) What hierarchy and dependence structure results from a data-adaptive threshold, and how sensitive is that structure to alternative analytical settings? (3) How do selected institutional and technological interventions behave when feedback effects and implementation delays are examined in an exploratory dynamic model? The study makes three contributions. First, it develops a reproducible 33-link direct-relation matrix from full-precision questionnaire means and treats unassessed relationships explicitly as instrument-defined structural non-links. This provides a transparent basis for identifying intervention priorities rather than relying on an unrestricted or retrospectively completed influence matrix. Second, it separates the conceptual organisation supplied by TOE from the causal, hierarchical, and dependence structures obtained through DEMATEL, ISM, and Matrix of Cross-Impact Multiplications Applied to Classification (MICMAC), while testing the stability of the findings through alternative coding, threshold sensitivity analysis, and bootstrap resampling. Third, it connects the structural results to an exploratory system dynamics model limited to factors represented in the Vensim equations. The resulting scenarios are used as policy experiments for comparing institutional, technological, and combined interventions rather than as forecasts of actual industry performance.

## 2 Conceptual Background and Factor Construction

### 2.1 Joint Institutional and Technological Management

The smart governance of engineering project credit risk is treated as a management system that connects institutional rules, participant responsibilities, project data, and supervisory action. Institutional arrangements specify who must provide guarantees, record defaults, verify performance, impose consequences, and authorise credit repair. Smart supervision technologies affect the speed, coverage, and consistency with which these obligations can be monitored and coordinated. A digital platform without enforceable responsibilities may simply reproduce existing ambiguities in electronic form, while rules unsupported by interoperable monitoring may remain slow and episodic in practice. The relevant unit of analysis is therefore the combined governance arrangement rather than a stand-alone technology. From a decision-making perspective, the central task is to determine which institutional and technological constraints exert the greatest influence and which interventions should receive priority when supervisory resources are limited.

Engineering project credit risk also develops across the project lifecycle. At the entry stage, decisions concern participant qualifications, performance guarantees, and permissions for accessing and sharing data. During contract performance, project participants assume obligations relating to payment, schedule, cost, quality, safety, and environmental compliance. Construction monitoring determines whether observed signals are verified and recorded as credit events. Default handling then connects the available evidence to sanctions, guarantee claims, corrective measures, appeals, and credit repair. This lifecycle perspective provides the basis for interpreting the causal structure and comparing the institutional and technological interventions after the structural and dynamic results have been obtained.

### 2.2 Position Relative to Integrated-Model Studies

Table 1 positions the present study as a decision-oriented integration of structural and dynamic analysis, with particular attention to traceability and clearly defined analytical roles. DEMATEL estimates factor prominence and net causal position, ISM arranges the retained relationships into hierarchical levels, and MICMAC classifies the factors according to their driving power and dependence. The SD model then examines the behaviour of selected feedback mechanisms under alternative intervention settings. These outputs address different parts of the decision problem and are used as complementary forms of evidence. They are not combined through an assumed one-to-one transfer of all 13 factors, and the DEMATEL prominence scores are not entered into the SD equations as numerical weights. The integration therefore supports intervention prioritisation without implying that the structural and dynamic models are mathematically identical.

**Table 1.** Focused comparison with related integrated-model studies

| Study                    | Method                            | Domain                                     | Distinguishing Feature                                                                                                                                                                                        |
|--------------------------|-----------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alsugair et al. [23]     | DEMATEL + SD                      | Project cost                               | Uses DEMATEL results to inform a dynamic model of project costs.                                                                                                                                              |
| Ajayi and Chinda [24]    | DEMATEL + SD                      | Schedule delay                             | Connects judgements about delay controls with schedule dynamics.                                                                                                                                              |
| Cheng et al. [25]        | DEMATEL + ISM + SD                | Smart-city emissions                       | Links the hierarchy of emission drivers to dynamic emission pathways.                                                                                                                                         |
| Rostamnezhad et al. [26] | Fuzzy DEMATEL + SD                | Construction social sustainability         | Combines influence judgements with the simulation of feedback relationships.                                                                                                                                  |
| Present study            | TOE + DEMATEL + ISM + MICMAC + SD | Engineering project credit-risk governance | Establishes a traceable path from a bounded 33-link instrument to factor prioritisation, hierarchical structuring, dependence classification, robustness assessment, and exploratory intervention comparison. |

Note: TOE = Technology–Organization–Environment; DEMATEL = Decision-Making Trial and Evaluation Laboratory; ISM = Interpretive Structural Modeling; MICMAC = Matrix of Cross-Impact Multiplications Applied to Classification; SD = System Dynamics.

### 2.3 Factor Development and Operational Boundaries

The final factor set was developed through an iterative review of the literature and policy documents, followed by the consolidation of overlapping concepts and refinement of the questionnaire. The available study records document the final 13 factors (S1–S13) and their wording but do not contain a recoverable count of the initial candidate factors, a formal deletion log, respondent demographic information, or a documented Delphi process. The study therefore reports only those factor-development procedures that can be verified from the available records. Under the TOE framework, S2 and S10 are assigned to Technology, S4, S5, S6, S7, S8, and S9 to Organization, and S1, S3, S11, S12, and S13 to Environment. S7–S9 are placed under Organization because they originate in project execution and the performance of participating organisations, even though their consequences may extend beyond organisational boundaries. These assignments provide a conceptual structure for the subsequent analysis, while the empirical models determine factor prominence, causal position, hierarchical level, dependence, and relevance to intervention decisions. The final factor system, TOE classification, and operational definitions are summarised in Table 2.

**Table 2.** TOE-organised factor system and operational definitions

| Code | Factor                                            | TOE          | Operational Definition                                                                                                          |
|------|---------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| S1   | Lack of credit guarantee mechanism                | Environment  | Missing project- or market-level performance guarantees, default-risk sharing arrangements, and enforceable credit constraints. |
| S2   | Backward smart supervision technology             | Technology   | Insufficient platform capability, data connectivity, real-time monitoring, or analytical support for credit supervision.        |
| S3   | Cross-subject credit information asymmetry        | Environment  | Fragmented or unequally accessible credit information among owners, contractors, supervision organisations, and regulators.     |
| S4   | Deviant credit behavior of construction units     | Organization | Owner or developer conduct that violates contractual commitments, payment duties, or regulatory requirements.                   |
| S5   | Deviant credit behavior of contractors            | Organization | Contractor conduct that violates contract, quality, safety, schedule, payment, or reporting obligations.                        |
| S6   | Lack of credit verification by supervision units  | Organization | Insufficient verification, recording, escalation, or follow-up of participant credit events by supervision organisations.       |
| S7   | Quality and safety credit default risk            | Organization | Quality and safety non-compliance generated within project execution and attributable to participant performance.               |
| S8   | Schedule and cost credit default risk             | Organization | Delay, overrun, payment, or cost-control failures arising during contractual delivery.                                          |
| S9   | Environmental and resource compliance credit risk | Organization | Failure to meet environmental, resource-use, or related compliance obligations during project delivery.                         |
| S10  | Lagging credit management methods                 | Technology   | Manual, fragmented, or delayed credit-management procedures and workflows, distinct from the digital capability itself.         |
| S11  | Absence of credit regulatory regulations          | Environment  | Gaps in external legal rules, regulatory standards, and statutory enforcement applicable to engineering credit.                 |
| S12  | Lack of integrity culture                         | Environment  | Weak shared norms and expectations supporting honest performance and credible reporting.                                        |
| S13  | Blocked social credit supervision channels        | Environment  | Limited public or third-party channels for reporting, accessing, and following up engineering credit information.               |

Note: TOE = Technology–Organization–Environment; S1–S13 denote the 13 engineering project credit-risk factors defined in the table.

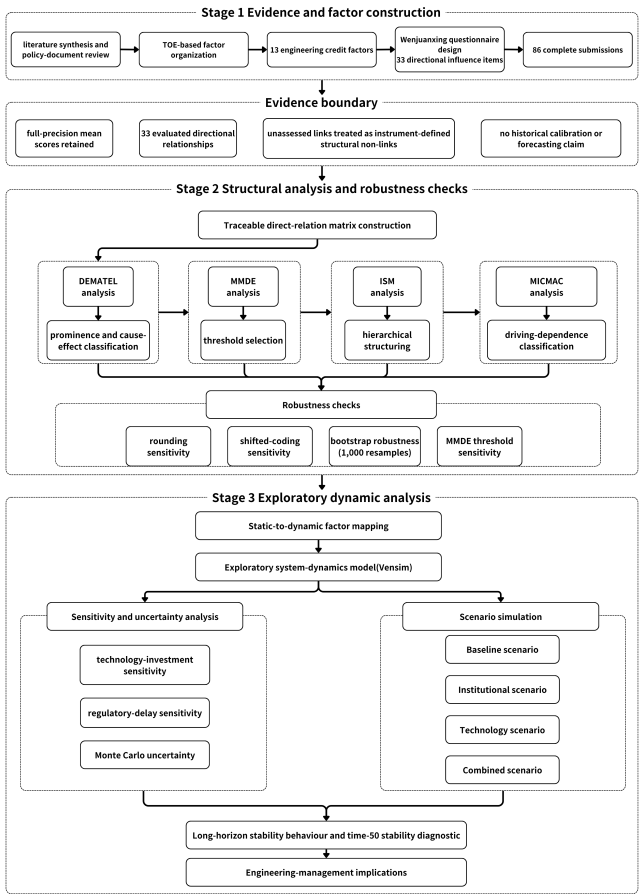
Several conceptual boundaries are important when interpreting the factor system. S1 concerns project- or market-level assurance arrangements, including performance guarantees, default-risk sharing, and the financial or contractual cost of non-performance. S11 instead concerns the external legal and regulatory framework that defines applicable standards and enforcement authority. Similarly, S2 refers to the capability of the digital infrastructure, including platforms, connectivity, real-time monitoring, and analytical functions. S10 refers to the procedures and workflows through which credit information is collected, verified, escalated, and used. It is placed under Technology because the factor, as defined in the instrument, concerns the extent to which these workflows remain manual, fragmented, and delayed rather than the broader allocation of organisational responsibilities.

The TOE categories are used to organise the factors conceptually; they do not predetermine the empirical relationships among them. An environmental factor such as S11 may therefore appear as autonomous if it has limited driving power and dependence within the assessed network. Conversely, an organisational risk such as S9 may occupy a deep causal position when its relationships with other factors give it substantial driving power. These empirical positions do not contradict the TOE assignments and should not be altered to produce a more conventional pattern. Instead, they show how the factors function within the questionnaire-defined decision system examined in this study.

### 3 Materials and Methods

#### 3.1 Research Design

Figure 1 summarises the analytical design. The DEMATEL method was used to quantify the direct and propagated influences among the identified factors [42]. The Maximum Mean De-Entropy (MMDE) algorithm was then applied to determine the threshold for retaining relationships [43]. ISM was used to arrange the retained relationships into hierarchical levels, while the MICMAC was used to classify the factors according to their driving power and dependence. Finally, a scenario-based system dynamics model was developed using stock-flow and feedback principles to compare selected institutional-response and technological interventions [44–46].



**Figure 1.** Analytical workflow from questionnaire-defined relationships to factor prioritisation, structural classification, and exploratory intervention comparison

Note: TOE = Technology–Organization–Environment; DEMATEL = Decision-Making Trial and Evaluation Laboratory; MMDE = Maximum Mean De-Entropy; ISM = Interpretive Structural Modeling; MICMAC = Matrix of Cross-Impact Multiplications Applied to Classification.



The methods addressed successive parts of the decision problem. DEMATEL identified the factors requiring priority attention, ISM represented their hierarchical relationships, and MICMAC distinguished potential drivers from dependent outcomes. The system dynamics model then examined how selected intervention levers behaved over time under the stated equations and parameter settings. The structural and dynamic analyses were connected through an explicit factor-mapping procedure rather than by transferring all 13 factors directly into the simulation.

### 3.2 Questionnaire Administration and Available Respondent Information

The questionnaire was administered exclusively online through Questionnaire Star (Wenjuanxing), a web-based survey platform. The archived record contained 86 submissions and 33 directional influence items, although the number of questionnaires or invitations distributed could not be recovered. All 86 records were complete, none were excluded, and all were retained for analysis. No exclusion criteria were applied because the archive contained no verifiable record of mechanical-response screening. Each item used a four-level scale, where 1 represented slight influence, 2 moderate influence, 3 significant influence, and 4 very strong influence. The instrument did not include 0 or a separate “no direct influence” option.

The anonymous dataset did not retain information on occupation, years of experience, organisational affiliation, main project type, engineering credit-supervision experience, or geographic group. Respondent-profile subgroup analyses could therefore not be conducted. The verified questionnaire-administration information and the available sample characteristics are summarised in Table 3.

**Table 3.** Questionnaire administration and available respondent information

| Item                                   | Verified Value                                                                                                        |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Administration mode                    | Online survey via Questionnaire Star (Wenjuanxing)                                                                    |
| Questionnaires/invitations distributed | Not recoverable from the archived Wenjuanxing record                                                                  |
| Submissions received                   | 86                                                                                                                    |
| Submissions excluded                   | 0                                                                                                                     |
| Exclusion criteria applied             | None; all 86 archived submissions were complete, and no verifiable mechanical-response screening record was available |
| Final analytical sample                | 86                                                                                                                    |
| Directional influence items            | 33                                                                                                                    |
| Influence scale                        | 1–4; no zero/no-influence option                                                                                      |
| Missing item responses                 | 0                                                                                                                     |
| Respondent profile/geography           | Not collected in the anonymous survey dataset                                                                         |

The absence of profile variables prevents post hoc comparisons of DEMATEL rankings across respondent groups. Bootstrap analysis can assess the stability of the aggregate rankings under resampling, but it cannot replace comparisons between professional, organisational, experiential, or geographic groups. Future questionnaires should therefore collect non-identifying information on respondent roles and experience under an explicit consent and data-minimisation protocol.

Because respondent occupations and professional experience were not retained in the archived dataset, the sample cannot be retrospectively verified as an expert panel. The resulting influence scores are therefore interpreted as aggregate questionnaire judgements rather than as formally validated expert consensus.

### 3.3 Construction of the Traceable Direct-Relation Matrix

The questionnaire completed by the 86 respondents assessed 33 of the 156 possible off-diagonal ordered pairs among the 13 factors. The direct-relation matrix  $A$  was constructed from these assessed relationships. Each of the 33 corresponding cells was assigned the full-precision arithmetic mean of the 86 item scores. The remaining 123 off-diagonal cells were set to zero because these relationships had not been evaluated by the questionnaire. The archived study materials did not contain a formal record explaining why these 33 directional relationships were selected from the 156 possible ordered pairs. Accordingly, the present analysis does not retrospectively reconstruct a link-selection procedure that cannot be verified. The resulting causal network should therefore be interpreted as conditional on the coverage of the original questionnaire instrument.

These zero entries were treated as instrument-defined structural non-links for the purpose of matrix construction. They do not represent zero ratings selected by respondents, nor do they demonstrate that the corresponding substantive relationships are absent. This distinction preserves the boundary of the questionnaire-defined network and prevents unassessed relationships from being interpreted as observed judgements. The 33 directional relationships actually

evaluated by the questionnaire, together with their full-precision means and response distributions, are reported in Table 4.

**Table 4.** Traceable directional influence links evaluated by the questionnaire

| Item | Direction  | Mean     | Standard Deviation | $n(1)$ | $n(2)$ | $n(3)$ | $n(4)$ |
|------|------------|----------|--------------------|--------|--------|--------|--------|
| 1    | S1 to S9   | 3.453488 | 0.849318           | 6      | 2      | 25     | 53     |
| 2    | S1 to S10  | 3.302326 | 0.736762           | 3      | 5      | 41     | 37     |
| 3    | S2 to S6   | 3.232558 | 0.890134           | 7      | 5      | 35     | 39     |
| 4    | S2 to S7   | 3.395349 | 0.786340           | 4      | 4      | 32     | 46     |
| 5    | S2 to S13  | 3.209302 | 0.704878           | 3      | 5      | 49     | 29     |
| 6    | S3 to S13  | 3.267442 | 0.758085           | 4      | 4      | 43     | 35     |
| 7    | S4 to S3   | 3.360465 | 0.780841           | 4      | 4      | 35     | 43     |
| 8    | S4 to S5   | 3.267442 | 0.726384           | 3      | 5      | 44     | 34     |
| 9    | S4 to S10  | 3.267442 | 0.788512           | 5      | 3      | 42     | 36     |
| 10   | S6 to S1   | 3.255814 | 0.689178           | 2      | 6      | 46     | 32     |
| 11   | S6 to S2   | 3.279070 | 0.697072           | 2      | 6      | 44     | 34     |
| 12   | S6 to S3   | 3.406977 | 0.757362           | 3      | 5      | 32     | 46     |
| 13   | S6 to S4   | 3.244186 | 0.750467           | 4      | 4      | 45     | 33     |
| 14   | S6 to S5   | 3.337209 | 0.776273           | 4      | 4      | 37     | 41     |
| 15   | S6 to S7   | 3.186047 | 0.759437           | 5      | 3      | 49     | 29     |
| 16   | S6 to S10  | 3.348837 | 0.715664           | 2      | 6      | 38     | 40     |
| 17   | S7 to S8   | 3.290698 | 0.700693           | 2      | 6      | 43     | 35     |
| 18   | S7 to S10  | 3.302326 | 0.704102           | 2      | 6      | 42     | 36     |
| 19   | S7 to S12  | 3.255814 | 0.842769           | 7      | 1      | 41     | 37     |
| 20   | S8 to S1   | 3.337209 | 0.806014           | 5      | 3      | 36     | 42     |
| 21   | S8 to S2   | 3.325581 | 0.742680           | 3      | 5      | 39     | 39     |
| 22   | S8 to S5   | 3.325581 | 0.710292           | 2      | 6      | 40     | 38     |
| 23   | S8 to S10  | 3.302326 | 0.768034           | 4      | 4      | 40     | 38     |
| 24   | S9 to S2   | 3.197674 | 0.699912           | 3      | 5      | 50     | 28     |
| 25   | S9 to S4   | 3.290698 | 0.733505           | 3      | 5      | 42     | 36     |
| 26   | S9 to S6   | 3.348837 | 0.715664           | 2      | 6      | 38     | 40     |
| 27   | S9 to S8   | 3.302326 | 0.827040           | 6      | 2      | 38     | 40     |
| 28   | S10 to S1  | 3.348837 | 0.747819           | 3      | 5      | 37     | 41     |
| 29   | S10 to S5  | 3.337209 | 0.679281           | 1      | 7      | 40     | 38     |
| 30   | S11 to S4  | 3.395349 | 0.786340           | 4      | 4      | 32     | 46     |
| 31   | S12 to S9  | 3.325581 | 0.710292           | 2      | 6      | 40     | 38     |
| 32   | S12 to S11 | 3.279070 | 0.791888           | 5      | 3      | 41     | 37     |
| 33   | S13 to S7  | 3.279070 | 0.791888           | 5      | 3      | 41     | 37     |

Note:  $n(k)$  denotes the number of respondents assigning score  $k$ , where  $k = 1, 2, 3$ , or  $4$ . Scores 1–4 represent slight, moderate, significant, and very strong influence, respectively. Full-precision means were retained in all calculations; six decimal places are displayed.

Because the forced-choice scale did not include a no-influence option, every assessed relationship necessarily received a positive score. The resulting means should therefore not be interpreted as evidence of the frequency with which respondents perceived no direct relationship. Restricting the matrix to the 33 relationships actually assessed prevented the unasked ordered pairs from being assigned artificial positive values. At the same time, it made the resulting network dependent on the coverage of the original instrument. Both consequences are recognised as limitations of the analysis.

### 3.4 Full-Precision Decision-Making Trial and Evaluation Laboratory Calculation

Let  $a_{ij}$  denote the mean reported influence of factor  $i$  on factor  $j$  for an evaluated directional item, and let  $a_{ij} = 0$  for an unasked structural non-link. All calculations use float64 values before display rounding.

$$s = \max_i \sum_j a_{ij}, \quad B = \frac{A}{s} \quad (1)$$

where,  $A$  is the traceable direct-relation matrix,  $s$  is its maximum row sum (23.0581395349), and  $B$  is the normalised direct-relation matrix.

$$T = B(I - B)^{-1} \quad (2)$$

where,  $T$  is the total-relation matrix and  $I$  is the  $13 \times 13$  identity matrix. The spectral radius of  $B$  is 0.3410762589, so the inverse exists for the observed matrix.

$$D_i = \sum_j t_{ij}, \quad R_i = \sum_j t_{ji} \quad (3)$$

where,  $t_{ij}$  is an element of  $T$ ,  $D_i$  is the total influence exerted by factor  $i$ , and  $R_i$  is the total influence received by factor  $i$ .

$$C_i = D_i + R_i, \quad N_i = D_i - R_i \quad (4)$$

where,  $C_i$  is prominence (centrality) and  $N_i$  is net relation.  $N_i > 0$  identifies a net cause and  $N_i < 0$  a net effect.

All substantive analysis was based on the full-precision traceable matrix, and values were rounded only for display. To distinguish the effect of numerical rounding from changes in network topology, the same sparse 33-link matrix was recalculated after the 33 means had been conventionally rounded to one decimal place. The legacy dense matrix was not used as the rounding comparator.

### 3.5 Maximum Mean De-Entropy Threshold, Interpretive Structural Modeling, and Matrix of Cross-Impact Multiplications Applied to Classification

The MMDE procedure is applied to the non-diagonal total-relation values and selects  $\lambda = 0.168627236856$ . A directed edge is retained when  $t_{ij}$  is at least  $\lambda$ . The diagonal is set to one for reachability analysis, and transitive closure yields the reachability matrix  $M$ . For factor  $i$ , the reachability set contains factors reachable from  $i$ , while the antecedent set contains factors that can reach  $i$ . A factor enters the current surface level when the intersection of those sets equals its reachability set; identified factors are removed iteratively until all levels are assigned.

$$m_{ij} = \begin{cases} 1, & t_{ij} \geq \lambda \\ 0, & t_{ij} < \lambda \end{cases} \quad (i \neq j) \quad (5)$$

where,  $m_{ij}$  is an off-diagonal element of the thresholded adjacency matrix and  $\lambda$  is the MMDE threshold. Diagonal elements are set to one before transitive closure.

$$P_i = \sum_j m_{ij}, \quad Q_i = \sum_j m_{ji} \quad (6)$$

where,  $P_i$  is MICMAC driving power and  $Q_i$  is dependence power in the final reachability matrix. The common classification cutoff is their mean, 2.461538.

### 3.6 Robustness Procedures

Four checks were conducted to assess measurement and model sensitivity. First, the same 33-link topology was recalculated using means rounded to one decimal place rather than full-precision means. The analysis considered both a newly estimated MMDE threshold and the fixed threshold obtained from the full-precision matrix.

Second, the original 1–4 scores were retained as the primary coding, while an alternative coding test mapped scores of 1, 2, 3, and 4 to 0, 1, 2, and 3, respectively. This recoding was used only as a sensitivity test and does not imply that the original questionnaire contained a zero or no-influence response option.

Third, 1,000 bootstrap samples of the 86 questionnaire records were drawn with replacement using seed 12202. DEMATEL prominence rankings and cause-effect classifications were recalculated for every bootstrap sample. This procedure examined whether the principal factor rankings depended on particular combinations of respondents within the available sample.

Fourth, the ISM structure was evaluated at MMDE – 10%, the selected MMDE threshold, MMDE + 10%, and the absolute thresholds of 0.38, 0.40, and 0.42. The absolute thresholds provided a scale-stress test and were deliberately high relative to the values in the sparse total-relation matrix. Together, these checks examined the stability of factor prioritisation, causal classification, and hierarchical structure under alternative numerical treatments.

### 3.7 Exploratory System Dynamics Model and Intervention Scenarios

The Vensim .mdl model defined a stock variable named the industry credit index and a set of feedback relationships involving regulatory pressure, guarantee completeness, smart supervision technology, information symmetry, quality and safety default risk, dishonesty cost, and the rates of credit improvement and decline. The industry credit index was treated as a dimensionless, model-generated representation of engineering-sector credit performance rather than an indicator estimated from observed industry data.

The equations were simulated using Euler integration with a time step of 0.25 over a simulation horizon of 0–50. One unit represented a simulation or policy interval and was not assumed to correspond to a calendar year. Times 5,



10, and 20 were selected as practical comparison points. Times 30–50 were used only to examine the long-horizon behaviour of the modelled trajectories and were not interpreted as forecasts of actual industry conditions.

Four scenarios were examined using the same model structure and equations. The baseline scenario retained the original parameter values. The institutional-response intervention reduced the regulatory response delay from 2.0 to 1.0. The technology intervention increased the technology investment coefficient from 1.0 to 2.0. The combined intervention applied both parameter changes simultaneously. Referring to the second scenario as an institutional-response intervention reflects its specific modelled mechanism and avoids implying that the simulation represents a comprehensive reform of the wider institutional system.

The scenarios were designed to compare the relative dynamic behaviour of selected governance levers under the stated model assumptions. The model was not fitted to historical observations of the engineering-sector credit index and should therefore be treated as an exploratory, scenario-based policy model. Learning-by-doing, diffusion, and feedback modelling supplied the conceptual basis for the dynamic relationships [44–46], but they did not convert the assumed equations or parameter values into empirically observed relationships. Accordingly, the scenario results were used to examine intervention timing, relative performance, and trajectory stability rather than to predict future industry credit levels.

## 4 Structural Results

### 4.1 Decision-Making Trial and Evaluation Laboratory Prominence and Net Relation

Table 5 reports the full-precision DEMATEL results, including prominence, net relation, cause–effect classification, ranking stability under one-decimal rounding, and bootstrap rank intervals.

**Table 5.** Full-precision DEMATEL results and rounding-only rank comparison

| Factor | D     | R     | D + R | D – R  | Group  | Full-Precision Rank | One-Decimal Rank* | Bootstrap 95% Rank Interval |
|--------|-------|-------|-------|--------|--------|---------------------|-------------------|-----------------------------|
| S6     | 1.420 | 0.434 | 1.854 | 0.986  | Cause  | 1                   | 1                 | 1–1                         |
| S9     | 1.069 | 0.435 | 1.504 | 0.634  | Cause  | 2                   | 2                 | 2–3                         |
| S10    | 0.363 | 1.119 | 1.482 | –0.756 | Effect | 3                   | 3                 | 2–3                         |
| S2     | 0.756 | 0.611 | 1.367 | 0.145  | Cause  | 4                   | 4                 | 4–4                         |
| S7     | 0.662 | 0.638 | 1.300 | 0.023  | Cause  | 5                   | 5                 | 5–5                         |
| S8     | 0.811 | 0.439 | 1.250 | 0.371  | Cause  | 6                   | 6                 | 6–6                         |
| S1     | 0.505 | 0.719 | 1.224 | –0.213 | Effect | 7                   | 7                 | 7–7                         |
| S4     | 0.506 | 0.580 | 1.086 | –0.073 | Effect | 8                   | 8                 | 8–8                         |
| S5     | 0.000 | 0.946 | 0.946 | –0.946 | Effect | 9                   | 9                 | 9–9                         |
| S12    | 0.472 | 0.231 | 0.704 | 0.241  | Cause  | 10                  | 10                | 10–10                       |
| S13    | 0.236 | 0.429 | 0.665 | –0.192 | Effect | 11                  | 11                | 11–11                       |
| S3     | 0.175 | 0.442 | 0.617 | –0.267 | Effect | 12                  | 12                | 12–12                       |
| S11    | 0.222 | 0.175 | 0.397 | 0.047  | Cause  | 13                  | 13                | 13–13                       |

Note: \*One-decimal ranks use the same 33 assessed links and conventional one-decimal rounding. Spearman  $\rho = 1.000$ ; no factor changes rank or cause/effect group. All inference uses the full-precision matrix. DEMATEL = Decision-Making Trial and Evaluation Laboratory; D = total influence exerted by a factor; R = total influence received by a factor; D + R = prominence; D – R = net relation;  $\rho$  = Spearman's rank correlation coefficient. A positive D – R value indicates a net cause, whereas a negative value indicates a net effect.

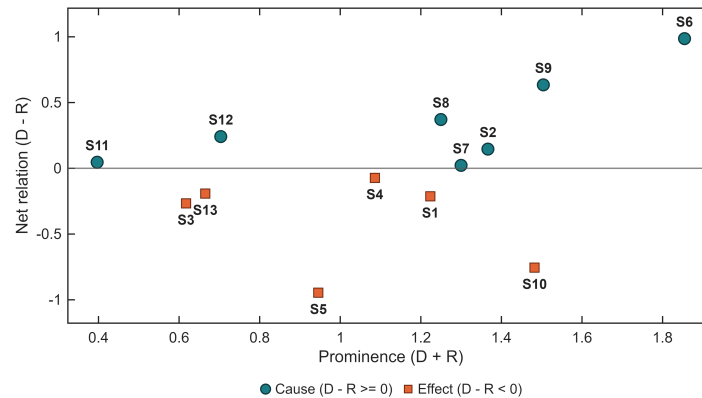
The full-precision prominence order was  $S6 > S9 > S10 > S2 > S7 > S8 > S1 > S4 > S5 > S12 > S13 > S3 > S11$ . S6 had the highest prominence value (1.854) and the largest positive net relation (0.986), identifying insufficient credit verification by supervision units as the most prominent net driver within the assessed network. S9 ranked second in prominence and was also classified as a net cause. S10 ranked third but had a strongly negative net relation (–0.756), indicating that lagging credit-management methods primarily received propagated influence from other factors.

The combination of high prominence and positive net influence placed S6 and S9 at the head of the initial intervention-priority set. This structural position does not by itself demonstrate that an intervention targeting either factor will produce the largest practical effect. The dynamic implications of selected intervention levers were therefore examined separately in the exploratory system dynamics model. All subsequent structural interpretation was based on the full-precision ordering.

The rounding-only analysis preserved all 13 prominence ranks, with Spearman's  $\rho = 1.000$ , and retained every cause-effect assignment. Re-estimating MMDE from the one-decimal matrix increased the threshold from 0.168627 to 0.183536, reduced the number of retained edges from 13 to 5, and produced a three-level hierarchy in which S7 and S8 moved to the surface level. When the full-precision MMDE threshold was held constant, the one-decimal matrix retained 13 edges and reproduced the original four-level hierarchy. The prominence rankings and net-relation

signs were therefore insensitive to rounding. The change in the hierarchy resulted from re-estimation of the threshold rather than from any change in the underlying 33-link topology.

The prominence and net-relation positions of all 13 factors are visualised in Figure 2.



**Figure 2.** DEMATEL prominence and net-relation positions based on the traceable full-precision matrix

Note: DEMATEL = Decision-Making Trial and Evaluation Laboratory; D = total influence exerted by a factor; R = total influence received by a factor; D + R = prominence; D - R = net relation. S1–S13 denote the 13 engineering project credit-risk factors.

Seven factors were classified as net causes: S2, S6, S7, S8, S9, S11, and S12. The remaining six factors—S1, S3, S4, S5, S10, and S13—were classified as net effects. A cause designation refers to the factor's net position in the assessed influence network; it does not imply moral responsibility, universal causal primacy, or proven intervention effectiveness. S11, for example, had a slightly positive net relation but the lowest prominence of all 13 factors. Its cause status must therefore be interpreted together with its limited connectivity in the threshold and MICMAC results and should not be used alone to support broad regulatory claims.

## 4.2 Threshold Sensitivity and Retained Relations

The effects of threshold selection and numerical rounding on the ISM structure are summarised in Table 6.

**Table 6.** ISM threshold and rounding sensitivity

| Threshold          | Value    | Edges | Isol. $n$ | Isolated Factors              | Levels | Surface-to-Root Partition                                    |
|--------------------|----------|-------|-----------|-------------------------------|--------|--------------------------------------------------------------|
| MMDE –10%          | 0.151765 | 19    | 3         | S11, S12, S13                 | 4      | L1:S3,S5,S10,S11,S12,S13   L2:S4,S7   L3:S2   L4:S1,S6,S8,S9 |
| MMDE               | 0.168627 | 13    | 3         | S11, S12, S13                 | 4      | L1:S1,S3,S4,S5,S10,S11,S12,S13   L2:S7,S8   L3:S2,S6   L4:S9 |
| 1 d.p., own MMDE   | 0.183536 | 5     | 6         | S3, S4, S8, S11, S12, S13     | 3      | L1:S1,S3,S4,S5,S7,S8,S10,S11,S12,S13   L2:S2,S6   L3:S9      |
| 1 d.p., fixed MMDE | 0.168627 | 13    | 3         | S11, S12, S13                 | 4      | L1:S1,S3,S4,S5,S10,S11,S12,S13   L2:S7,S8   L3:S2,S6   L4:S9 |
| MMDE +10%          | 0.185490 | 4     | 7         | S1, S3, S4, S8, S11, S12, S13 | 3      | L1:S1,S3,S4,S5,S7,S8,S10,S11,S12,S13   L2:S2,S6   L3:S9      |
| Absolute 0.38      | 0.380000 | 0     | 13        | S1–S13                        | 1      | L1:S1–S13                                                    |
| Absolute 0.40      | 0.400000 | 0     | 13        | S1–S13                        | 1      | L1:S1–S13                                                    |
| Absolute 0.42      | 0.420000 | 0     | 13        | S1–S13                        | 1      | L1:S1–S13                                                    |

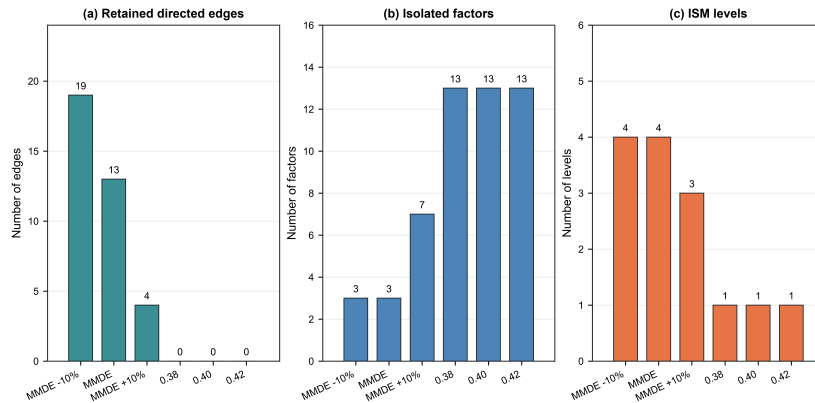
Note: The one-decimal rows retain the same 33-link topology and differ only in numerical precision and threshold treatment. The absolute thresholds 0.38, 0.40, and 0.42 are high relative to the sparse total-relation matrix and retain no off-diagonal edges. Isol.  $n$  = number of isolated factors; MMDE = Maximum Mean De-Entropy; ISM = Interpretive Structural Modeling; d.p. = decimal place; L1–L4 denote Levels 1–4 from surface to root; S1–S13 denote the 13 engineering project credit-risk factors.

At the selected MMDE threshold, 13 relationships were retained. The four strongest were S6 to S10 (0.2248), S6 to S5 (0.2064), S2 to S7 (0.1919), and S9 to S2 (0.1867). These relationships connected supervision verification, smart supervision technology, execution risk, and credit-management procedures within the main structural pathway.

At MMDE – 10%, 19 relationships remained, while S11, S12, and S13 were isolated. At MMDE + 10%, only four relationships remained and seven factors were isolated. S11, S12, and S13 remained isolated under every tested threshold. The three absolute thresholds removed all off-diagonal relationships and consequently placed every factor in a single surface level.

Under the formal ISM procedure, an isolated node is assigned to the surface partition because its reachability set contains only itself. A Level 1 position therefore does not necessarily identify an operational outcome, a highly dependent factor, or a factor of little managerial importance. It only describes the position produced by the retained relationships in the questionnaire-defined network. The rapid reduction in retained edges at higher thresholds also shows that the detailed hierarchy was more sensitive to threshold selection than the DEMATEL prominence ranking.

Figure 3 further illustrates how the number of retained edges, isolated factors, and ISM levels changed across the tested thresholds.



**Figure 3.** Threshold sensitivity of retained relationships, isolated factors, and ISM levels

Note: MMDE = Maximum Mean De-Entropy; ISM = Interpretive Structural Modeling. The number of ISM levels indicates the hierarchical depth obtained under each tested threshold.

#### 4.3 Interpretive Structural Modeling Hierarchy and Matrix of Cross-Impact Multiplications Applied to Classification

Table 7 summarises the ISM level, driving power, dependence power, and MICMAC classification of each factor.

**Table 7.** ISM level and MICMAC position of each factor

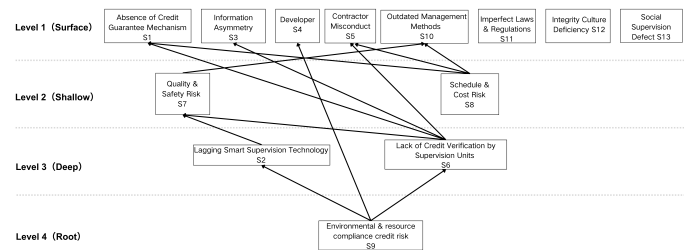
| Code | Factor                                            | TOE          | ISM     | Driving | Depend. | MICMAC Class |
|------|---------------------------------------------------|--------------|---------|---------|---------|--------------|
| S1   | Lack of credit guarantee mechanism                | Environment  | Level 1 | 1       | 4       | Dependent    |
| S2   | Backward smart supervision technology             | Technology   | Level 3 | 3       | 2       | Independent  |
| S3   | Cross-subject credit information asymmetry        | Environment  | Level 1 | 1       | 3       | Dependent    |
| S4   | Deviant credit behavior of construction units     | Organization | Level 1 | 1       | 2       | Autonomous   |
| S5   | Deviant credit behavior of contractors            | Organization | Level 1 | 1       | 4       | Dependent    |
| S6   | Lack of credit verification by supervision units  | Organization | Level 3 | 6       | 2       | Independent  |
| S7   | Quality and safety credit default risk            | Organization | Level 2 | 2       | 4       | Dependent    |
| S8   | Schedule and cost credit default risk             | Organization | Level 2 | 4       | 1       | Independent  |
| S9   | Environmental and resource compliance credit risk | Organization | Level 4 | 9       | 1       | Independent  |
| S10  | Lagging credit management methods                 | Technology   | Level 1 | 1       | 6       | Dependent    |
| S11  | Absence of credit regulatory regulations          | Environment  | Level 1 | 1       | 1       | Autonomous   |
| S12  | Lack of integrity culture                         | Environment  | Level 1 | 1       | 1       | Autonomous   |
| S13  | Blocked social credit supervision channels        | Environment  | Level 1 | 1       | 1       | Autonomous   |

Note: TOE = Technology–Organization–Environment; ISM = Interpretive Structural Modeling; MICMAC = Matrix of Cross-Impact Multiplications Applied to Classification; S1–S13 denote the 13 engineering project credit-risk factors.

The ISM analysis produced four levels extending from surface to root. Level 1 contained S1, S3, S4, S5, S10, S11, S12, and S13. Level 2 contained S7 and S8, while Level 3 contained S2 and S6. S9 was the sole factor at Level 4.

Within the assessed network, the hierarchy indicated a pathway in which environmental and resource compliance credit risk was connected to smart supervision technology and verification capacity. These factors were subsequently linked to project-execution risks and several observable governance deficiencies. The position of S9 at the root of the hierarchy applied specifically to the questionnaire-defined 33-link network. It should not be interpreted as evidence that S9 is the universal root cause of engineering project credit risk in all institutional or project settings.

Figure 4 presents the transitive reduction of the MMDE-thresholded directed acyclic graph rather than all 13 retained relationships. The direct S6-to-S10 edge was omitted because S6 and S10 were also connected through the S6-to-S7-to-S10 pathway. Removing this redundant edge left 12 irreducible relationships. Irreducible edges extending across non-adjacent levels were retained.



**Figure 4.** Four-level ISM hierarchy at the MMDE threshold after transitive reduction. Levels 1–4 represent surface, intermediate, deep, and root factors, respectively; only irreducible relationships are shown

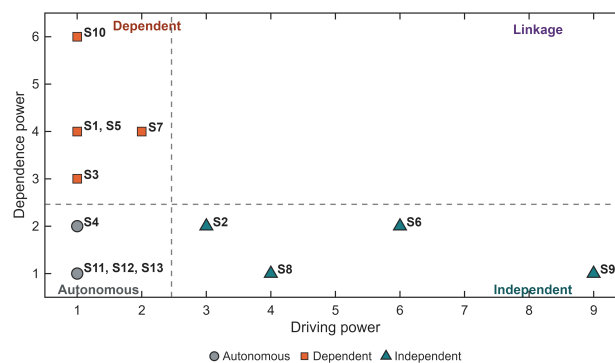
Note: ISM = Interpretive Structural Modeling; MMDE = Maximum Mean De-Entropy; S1–S13 denote the 13 engineering project credit-risk factors.

The MICMAC analysis classified S2, S6, S8, and S9 as independent factors; S1, S3, S5, S7, and S10 as dependent factors; and S4, S11, S12, and S13 as autonomous factors. No linkage factor appeared at the mean cutoff. In MICMAC terminology, “independent” refers to factors with relatively high driving power and low dependence; it does not mean statistical independence.

The combined ISM and MICMAC results identified S9 as both a root-level factor and an independent driver, while S6 occupied a deep level and also had substantial driving power. S2 and S8 were likewise classified as independent, although their prominence values were lower than those of S6 and S9. These differences show why factor priority should not be determined from a single indicator. Prominence describes overall involvement in the network, net relation distinguishes causes from effects, ISM represents hierarchical position, and MICMAC describes driving power and dependence.

S11 illustrates the distinction between conceptual classification and empirical connectivity. TOE classified S11 as an environmental condition because it concerns the external regulatory framework, whereas MICMAC classified it as autonomous because it had low driving power and low dependence in the assessed network. The two classifications therefore answer different questions and are not contradictory.

The resulting driving–dependence positions and MICMAC classifications are shown in Figure 5.



**Figure 5.** MICMAC driving-dependence classification at the MMDE threshold

Note: MICMAC = Matrix of Cross-Impact Multiplications Applied to Classification; MMDE = Maximum Mean De-Entropy; S1–S13 denote the 13 engineering project credit-risk factors. The dashed vertical and horizontal lines indicate the common mean cutoff of 2.461538 for driving power and dependence.

#### 4.4 Coding and Bootstrap Robustness

The alternative 0–3 coding produced the same prominence order as the primary 1–4 coding, with Spearman's  $\rho = 1.000$ . The factor ranking was therefore unchanged by a uniform one-point shift in the response coding. This test could not, however, reproduce a genuine no-influence response because such an option had not been included in the original questionnaire.

Across the 1,000 bootstrap resamples, S6 ranked first in every draw. S9 and S10 occupied ranks 2 and 3 in every draw, and each of the three factors appeared in the top three with a frequency of 1.000. Cause-effect membership remained unchanged for all factors except S7, which was classified as a net cause in 99.6% of the bootstrap draws.

These findings indicated that the principal prominence ordering was stable under resampling of the observed responses. The narrow rank intervals should nevertheless be interpreted in light of the fixed 33-link topology. The bootstrap procedure represented sampling variation in the collected item scores, but it did not capture uncertainty about the 123 directional relationships that were not assessed by the questionnaire. The robustness results therefore support the stability of factor prioritisation within the observed network rather than the completeness of the network itself.

### 5 Exploratory System Dynamics Simulation

#### 5.1 Static-to-Dynamic Mapping

Table 8 reports how each factor from the DEMATEL–ISM–MICMAC analysis was represented in the system dynamics model. S1–S7 and S11 were included either directly, through an explicitly identified composite variable, or as an exogenous model input. S8, S9, S10, S12, and S13 were not simulated because the source model contained no corresponding equations. The mapping did not transfer DEMATEL prominence values into the simulation as coefficients, nor did it assume that the static hierarchy had been dynamically validated. Instead, the system dynamics model examined a narrower set of feedback relationships associated with regulatory response and technology investment.

**Table 8.** Mapping from DEMATEL–ISM–MICMAC factors to system dynamics variables

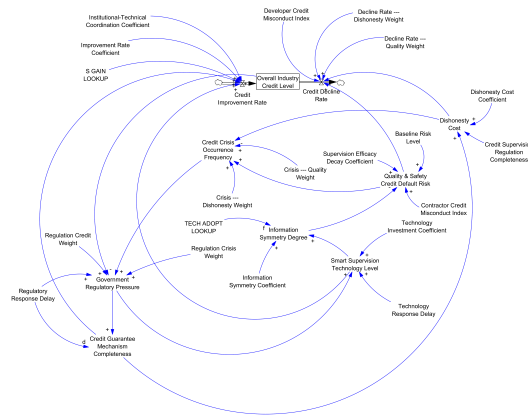
| Code | SD Variable                                    | Type                | Representation   | Sign | Equation/Parameter                         |
|------|------------------------------------------------|---------------------|------------------|------|--------------------------------------------|
| S1   | Credit guarantee completeness; dishonesty cost | Auxiliary/composite | Direct/composite | +    | Guarantee smooth; dishonesty-cost equation |
| S2   | Smart-supervision technology level             | Auxiliary           | Direct           | +    | Technology smooth; investment coefficient  |
| S3   | Information symmetry degree                    | Auxiliary           | Direct           | +    | Technology-adoption lookup                 |
| S4   | Developer misconduct index                     | Exogenous/step      | Direct           | –    | Credit-decline step at time 5              |
| S5   | Contractor misconduct index                    | Exogenous           | Direct           | –    | Quality/safety risk equation               |
| S6   | Supervision efficacy decay coefficient         | Parameter           | Composite proxy  | –    | Quality/safety risk equation               |
| S7   | Quality and safety default risk                | Auxiliary           | Direct           | –    | Risk and decline equations                 |
| S8   | None                                           | Not included        | Not simulated    | n/a  | No source-model equation                   |
| S9   | None                                           | Not included        | Not simulated    | n/a  | No source-model equation                   |
| S10  | None                                           | Not included        | Not simulated    | n/a  | No source-model equation                   |
| S11  | Regulation completeness                        | Exogenous           | Direct           | +    | Dishonesty-cost conditional                |
| S12  | None                                           | Not included        | Not simulated    | n/a  | No source-model equation                   |
| S13  | None                                           | Not included        | Not simulated    | n/a  | No source-model equation                   |

Note: DEMATEL = Decision-Making Trial and Evaluation Laboratory; ISM = Interpretive Structural Modeling; MICMAC = Matrix of Cross-Impact Multiplications Applied to Classification; SD = System Dynamics; n/a = not applicable. A positive sign indicates that an increase in the mapped governance capability raises the credit index, whereas a negative sign indicates that an increase in misconduct, risk, or decay lowers the index.

This distinction is important for interpreting the intervention results. S9 occupied the root position in the ISM

hierarchy, and S10 ranked third in DEMATEL prominence, but neither factor was represented in the dynamic model. S6 was represented only indirectly through a composite proxy for the decay of supervision efficacy. Accordingly, the simulation did not test whether interventions directed specifically at S9 or S10 would outperform the modelled technology-investment and regulatory-response interventions. It provided a separate exploratory comparison of the intervention levers available in the source model.

Figure 6 shows the stock-flow structure and the principal feedback relationships used in the exploratory system dynamics model.



Note: “+” and “-” denote positive and negative causal relationships, respectively; f denotes a functional relationship; d denotes a delay relationship. S GAIN LOOKUP is an S-shaped gain lookup function that converts the smart-supervision technology level into a gain coefficient, whereas TECH ADOPT LOOKUP is a technology-adoption lookup function that converts technology-related input into the information-symmetry degree.

The industry credit index,  $CI$ , was specified as a dimensionless endogenous stock with an initial value of 50. This value represented a reference state within the model rather than an observed or officially defined industry credit score. The stock was not bounded because the source equation integrated improvement minus decline without imposing a lower limit. A negative value therefore represented a modelled credit deficit relative to the reference state; it did not represent a negative administrative credit score. This formulation exposed the effect of adverse feedback within the model but restricted direct interpretation of the numerical values in real-world terms.

**Table 9.** Core system dynamics equations and variable explanations

Note:  $CI$  = credit index;  $INTEG$  = stock integration;  $STEP$  = step function;  $SMOOTH$  and  $SMOOTH1$  = first-order smoothing functions without and with an initial value, respectively;  $\min$  and  $\max$  denote minimum and maximum operators. Time  $t$  is measured in simulation intervals.



Table 10 reports the baseline values, uncertainty ranges, and provenance of the principal model parameters.

**Table 10.** Parameter provenance and uncertainty treatment

| Parameter                           | Baseline | Range/Treatment             | Basis                                  | Calibration Target    |
|-------------------------------------|----------|-----------------------------|----------------------------------------|-----------------------|
| Improvement rate coefficient        | 0.5      | $\pm 20\%$ MC               | Archived source-model value (retained) | No historical fit     |
| Institution-technology coordination | 1.0      | Fixed                       | Archived source-model value (retained) | No historical fit     |
| Decline quality weight              | 0.3      | $\pm 20\%$ MC               | Archived source-model value (retained) | No historical fit     |
| Decline dishonesty weight           | 0.2      | Fixed                       | Archived source-model value (retained) | No historical fit     |
| Regulatory response delay           | 2.0      | 1.6–2.4 scan; $\pm 20\%$ MC | Archived source-model value (retained) | No historical fit     |
| Technology response delay           | 1.5      | Fixed                       | Archived source-model value (retained) | No historical fit     |
| Regulation credit/crisis weights    | 0.6/0.4  | Fixed                       | Archived structural weights (retained) | No historical fit     |
| Information symmetry coefficient    | 0.7      | $\pm 20\%$ MC               | Archived source-model value (retained) | No historical fit     |
| Baseline risk level                 | 50       | Fixed                       | Archived reference value (retained)    | No historical fit     |
| Dishonesty cost coefficient         | 1.2      | Fixed                       | Archived source-model value (retained) | No historical fit     |
| Technology investment coefficient   | 1.0      | 0.8–1.2 scan; $\pm 20\%$ MC | Archived source-model value (retained) | No historical fit     |
| Crisis quality/default weights      | 0.4/0.3  | Fixed                       | Archived structural weights (retained) | No historical fit     |
| Regulation completeness             | 50       | Fixed                       | Archived reference value (retained)    | No historical fit     |
| Developer misconduct index          | 50       | Fixed; step at time 5       | Archived reference value (retained)    | No historical fit     |
| Contractor misconduct index         | 50       | Fixed                       | Archived reference value (retained)    | No historical fit     |
| Supervision efficacy decay          | 1.0      | Fixed                       | Archived reference value (retained)    | No historical fit     |
| Nonlinearity exponent               | 0.8      | Fixed                       | Archived source-model value (retained) | No historical fit     |
| Euler time step                     | 0.25     | Fixed                       | Archived Vensim setting (retained)     | Equation reproduction |

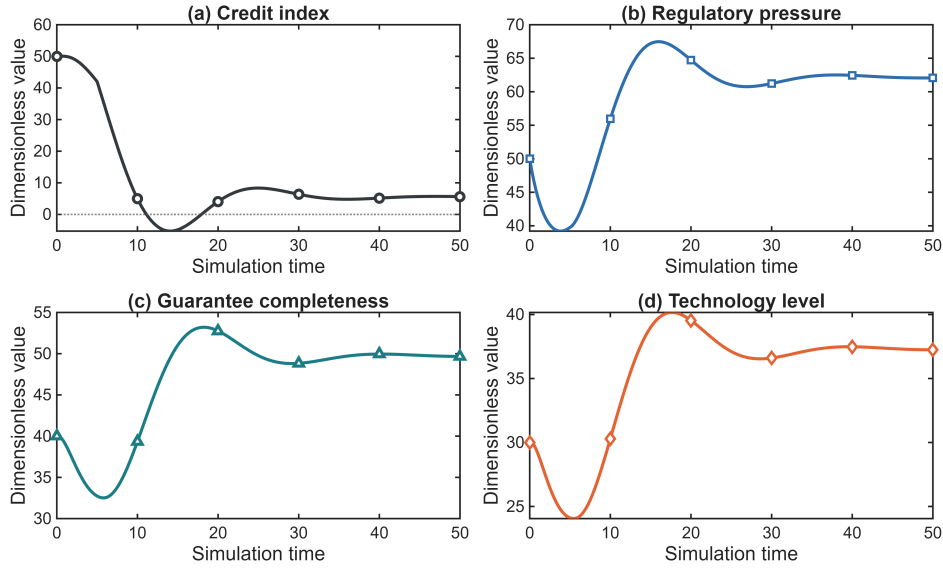
Note: MC = Monte Carlo. The archived source-model values, structural weights, reference values, and Vensim time step were retained unchanged to preserve reproducibility because no empirical calibration record was available. These parameters were not estimated from observed industry time-series data.

The equations were simulated using Euler integration with a time step of 0.25 over a horizon of 0–50. One unit represented a simulation or policy interval and was not assumed to correspond to a calendar year. Times 5, 10, and 20 were used as practical comparison points, while time 50 was used only as a long-horizon diagnostic. None of these points should be interpreted as a forecast date.

### 5.3 Baseline Behaviour

Under the baseline settings, *CI* started at 50 and decreased to 42.02 at time 5 and 4.98 at time 10. It reached its lowest reported value of  $-5.27$  at time 14 before recovering to 4.06 at time 20. The trajectory subsequently approached a low oscillatory regime, reaching 6.38 at time 30 and 5.64 at time 50.

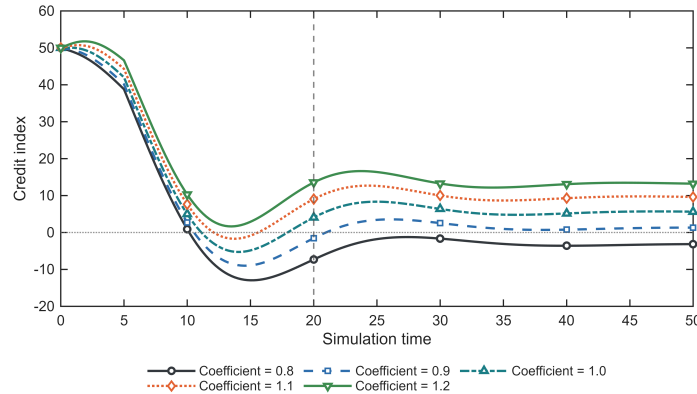
Figure 7 presents the corresponding trajectories of regulatory pressure, credit guarantee completeness, technology level, and *CI*. The temporary negative value resulted from the unbounded stock equation and represented a deficit relative to the model reference state. It should not be interpreted as an observed engineering-industry credit score.



**Figure 7.** Baseline trajectories of the credit index and principal feedback variables

#### 5.4 Parameter Sensitivity and Uncertainty

The technology-investment scan showed a monotonic pattern over the range of 0.8–1.2. At time 20, CI increased from  $-7.31$  at a coefficient of 0.8 to  $-1.54$  at 0.9, 4.06 at 1.0, 9.05 at 1.1, and 13.54 at 1.2. Within this local range, the technology-investment coefficient affected both the timing of recovery and the value reached at the later simulation checkpoint. The corresponding sensitivity trajectories are shown in Figure 8.

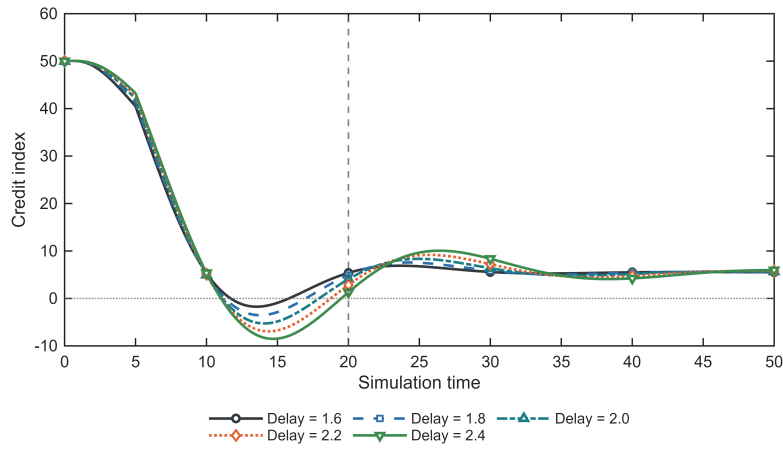


**Figure 8.** Sensitivity of the credit index to the technology investment coefficient

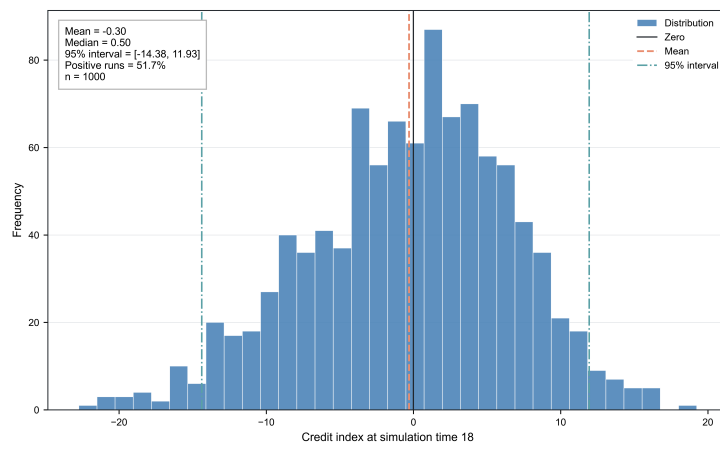
The regulatory-response-delay scan produced the opposite time-20 pattern. The corresponding  $CI$  values were 5.39, 4.91, 4.06, 2.84, and 1.29 for delays of 1.6, 1.8, 2.0, 2.2, and 2.4, respectively. A shorter response delay produced a higher time-20 value within this local scan. However, the transient trajectory was nonlinear, and the separate response-delay intervention did not produce a higher time-50 value than the baseline scenario. Figure 9 shows the corresponding sensitivity of  $CI$  to changes in regulatory response delay.

A 1,000-draw Monte Carlo analysis was conducted using independent triangular distributions for five parameters, with the lower bound, mode, and upper bound set at  $-20\%$ , the baseline value, and  $+20\%$ , respectively. At time 18, the mean  $CI$  was  $-0.30$  and the median was 0.50. The empirical 95% interval ranged from  $-14.38$  to 11.93, and 51.7% of the simulated values were positive. Because this interval crossed zero, the timing of the baseline recovery should not be treated as precise. The resulting distribution of  $CI$  at simulation time 18 is shown in Figure 10.

At time 18, the improvement-rate coefficient had the strongest rank correlation with  $CI$  ( $\rho = 0.702$ ), followed by the technology-investment coefficient ( $\rho = 0.560$ ). The remaining correlations were  $-0.224$  for the decline-quality weight,  $-0.172$  for regulatory response delay, and 0.137 for the information-symmetry coefficient. These values describe model sensitivity under the stated uncertainty design and are not empirical estimates of policy or causal effects. Figure 11 presents the Monte Carlo trajectory envelope together with the rank-based parameter sensitivity results.

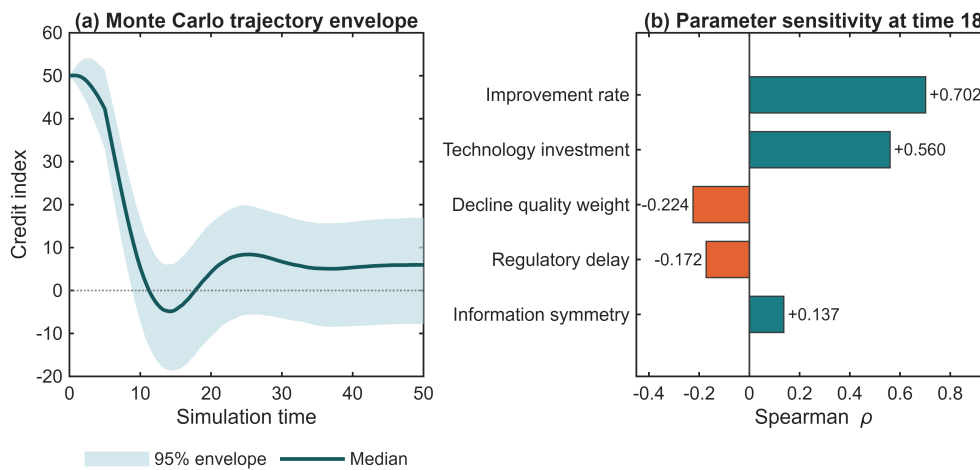


**Figure 9.** Sensitivity of the credit index to regulatory response delay



**Figure 10.** Monte Carlo distribution of the dimensionless credit index at simulation time 18

Note:  $n = 1,000$  denotes the number of Monte Carlo simulation runs; the 95% interval represents the empirical 2.5th–97.5th percentile interval.



**Figure 11.** Monte Carlo trajectory envelope and rank-based parameter sensitivity

Note:  $\rho$  denotes Spearman's rank correlation coefficient.

## 5.5 Scenario Comparison

Four scenarios were compared using the same equations and initial conditions. The baseline scenario retained all original parameter values. The institutional-response intervention reduced regulatory response delay from 2.0 to 1.0. The technology intervention increased the technology-investment coefficient from 1.0 to 2.0. The combined intervention applied both parameter changes.

The technology-investment value of 2.0 represented a discrete policy experiment and lay outside the local sensitivity range of 0.8–1.2 examined in Section 5.4. The local scan assessed behaviour around the baseline parameter, whereas the scenario analysis examined the model response to a larger hypothetical change. The results of the technology scenario should therefore be interpreted as stress-test evidence under the model equations rather than as an empirically calibrated estimate of a feasible investment programme.

The credit-index values and differences from baseline at the selected comparison points are summarised in Table 11.

**Table 11.** Credit-index scenarios and differences from baseline

| Scenario      | $t = 5$ | Delta  | $t = 10$ | Delta  | $t = 20$ | Delta  | $t = 50^*$ | Delta  |
|---------------|---------|--------|----------|--------|----------|--------|------------|--------|
| Baseline      | 42.02   | 0.00   | 4.98     | 0.00   | 4.06     | 0.00   | 5.64       | 0.00   |
| Institutional | 37.69   | −4.32  | 7.73     | +2.75  | 5.43     | +1.37  | 5.52       | −0.13  |
| Technology    | 67.61   | +25.59 | 25.85    | +20.86 | 39.09    | +35.03 | 33.42      | +27.77 |
| Combined      | 56.54   | +14.53 | 28.78    | +23.79 | 34.22    | +30.16 | 33.73      | +28.09 |

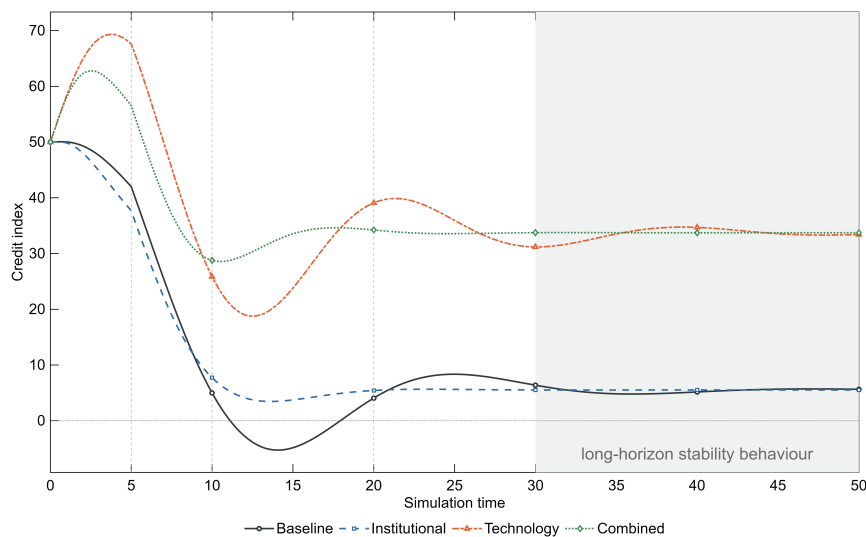
Note: Delta denotes the difference between each intervention scenario and the baseline at the same simulation time;  $t$  = simulation time. \*Time 50 is a long-horizon model diagnostic, not a forecast.

The technology intervention produced the highest *CI* values at times 5 and 20. Under the model equations, doubling the investment coefficient directly increased the technology stock, information symmetry, and the technology-related contribution to credit improvement. The combined intervention produced the highest value at time 10 but remained below the technology-only scenario at time 20.

At time 50, the combined scenario produced a value of 33.73, compared with 33.42 under the technology-only scenario. The difference of 0.31 was small and should not be interpreted as evidence that the combined intervention was materially more stable or effective. It only shows that the combined scenario produced a slightly higher value at this long-horizon diagnostic point.

The institutional-response intervention produced a value 4.32 below the baseline at time 5, 2.75 above the baseline at time 10, and 1.37 above the baseline at time 20. At time 50, it was 0.13 below the baseline. Reducing the response delay changed the timing of regulatory pressure and guarantee adjustment but did not introduce a new long-run target or a separate enforcement-capacity stock. Under the stated parameterisation, the institutional-response intervention moderated the downturn and improved selected intermediate outcomes but did not produce a clear long-run advantage.

These comparisons indicate that the relative performance of the modelled interventions depended on both the intervention mechanism and the selected time point. They do not establish that technology investment is universally preferable to institutional reform, particularly because the institutional-response scenario represented only a change in response delay and the model did not explicitly include S8, S9, S10, S12, or S13. The scenario results therefore support conditional intervention comparison within the stated model rather than a general ranking of all available engineering project credit-risk policies. The trajectories of the four scenarios are shown in Figure 12.



**Figure 12.** Credit-index trajectories under the baseline, institutional-response, technology, and combined scenarios

## 6 Discussion

### 6.1 Joint Interpretation of the Structural and Dynamic Analyses

The structural and dynamic analyses addressed related aspects of engineering project credit-risk governance, but they did not operate with identical variables or provide interchangeable evidence. In the structural analysis, S6 was the most prominent factor and had the largest positive net relation. This result indicates that supervision organisations occupy a central position in the assessed network because project information becomes usable for governance only after relevant events have been verified, recorded, and escalated. S9 occupied the root level of the ISM hierarchy and had the highest MICMAC driving power. Within the questionnaire-defined network, environmental and resource compliance risk therefore functioned as an upstream condition connecting project data, supervisory capacity, and organisational obligations. S10 was prominent but dependent, suggesting that improvements in credit-management procedures may rely on prior changes in verification, technological capability, and the project-execution controls that supply information to those procedures.

The dynamic analysis examined a narrower set of mechanisms. Under the selected scenario settings, the technology intervention produced higher credit-index values than the institutional-response intervention at times 5 and 20. This pattern arose because technology investment entered two positive pathways in the model: it increased information symmetry, which reduced quality risk, and it increased the technology-related contribution to credit improvement. The institutional-response intervention changed the regulatory response delay but left the long-run target structure, regulation-completeness reference, and enforcement-capacity structure unchanged. A shorter delay could therefore alter the timing of regulatory pressure and guarantee adjustment without necessarily producing a higher long-horizon value.

These scenario differences should not be interpreted as evidence that technology investment is generally more effective than institutional reform. The two interventions changed different parameters by different magnitudes, and the technology scenario used a coefficient of 2.0, which lay outside the local sensitivity range of 0.8–1.2. In addition, S9 and S10 were not explicitly included in the system dynamics model, while S6 was represented only through a composite proxy. The simulation therefore compared the available levers under the stated equations rather than validating the complete structural ranking.

At time 50, the combined scenario produced a credit-index value of 33.73, compared with 33.42 under the technology-only scenario. This difference of 0.31 was too small to establish that the combined intervention was materially more stable or more effective. It only indicates that the combined setting produced a slightly higher value at the selected long-horizon diagnostic point. Demonstrating greater stability would require separate evidence on convergence, oscillation amplitude, variance, or another defined stability measure.

### 6.2 Engineering-Management Actions Across the Project Lifecycle

At project entry, regulators and project owners should establish qualification requirements, performance guarantees, shared identifiers, and data-access permissions before contract award. These actions relate primarily to S1, S3, S11, and S13. The required management output is not simply a connected database. It is an auditable entry decision that identifies the responsible organisation, the applicable guarantee instrument, the authority to access and verify data, the conditions that trigger further action, and the available appeal route.

During contract performance, project owners and contractors should translate the risks represented by S4, S5, and S8 into measurable obligations concerning payment, schedule, cost, disclosure, and change control. Credit consequences should be linked to verified events and proportionate remedies. This distinction is necessary because delays, cost variations, and contractual changes do not automatically constitute misconduct. A documented assessment process is required to distinguish an ordinary project variation from material non-performance and to connect a verified default with guarantee activation, corrective action, or subsequent credit repair.

During construction monitoring, supervision organisations and regulators should coordinate the factors represented by S2, S6, S7, S9, and S10. A smart supervision platform should retain source evidence and provide traceable records, but the supervision organisation remains responsible for verification, exception classification, and escalation. Building information modelling, connected sensors, distributed ledgers, and analytical systems may support these tasks where their functions match the required evidence chain. However, the present model did not estimate the separate effect of any individual technology and does not support a ranking of specific technological solutions.

Environmental and resource compliance requires particular attention because S9 occupied the root of the ISM hierarchy and had the highest MICMAC driving power. In operational terms, environmental and resource obligations should be converted into measurable project requirements, assigned to responsible participants, monitored through verifiable records, and connected to defined responses. This recommendation follows from the structural analysis rather than the system dynamics model because S9 was not explicitly simulated.

During default handling, regulators, project owners, contractors, supervision organisations, and guarantee providers should connect the verified event record to notification, correction, sanction, guarantee claims, appeals, and

credit repair. S1 supplies the project- or market-level assurance mechanism, S11 represents the external regulatory setting, and S3 and S13 affect access to credit information. The autonomous MICMAC positions of S11 and S13 in this dataset call for careful interpretation. Their inclusion in governance arrangements is supported by their legal and operational functions rather than by strong connectivity in the assessed network.

### **6.3 Interpreting Technology–Organization–Environment Categories, Isolated Factors, and the Interpretive Structural Modeling Hierarchy**

The empirical hierarchy did not reproduce the TOE categories, nor was it expected to do so. TOE describes the nature of each condition, whereas DEMATEL, ISM, and MICMAC describe how the factor was positioned within the assessed relationship network. S7–S9 remained organisational factors because they arose from project execution and participant performance. S11 remained an environmental factor because it concerned external laws, regulations, and enforcement authority, even though it had low prominence, was isolated at the selected MMDE threshold, and was classified as autonomous by MICMAC.

These results demonstrate why conceptual classification and empirical network position should be interpreted separately. A factor may be institutionally important while having limited connectivity in a particular questionnaire-defined network. Conversely, an organisational risk may occupy a root position when the assessed relationships give it substantial driving power. Neither result requires the factor to be reassigned to another TOE category.

The absolute thresholds of 0.38–0.42 retained no off-diagonal edges in the sparse matrix. This was a consequence of scale and sparsity rather than evidence that all 13 factors were substantively independent. The local MMDE sensitivity analysis was more informative for evaluating the selected hierarchy. It showed that the central pathway connecting S9 with S2 and S6 persisted around the selected threshold, while the placement of peripheral factors was more sensitive to small threshold changes.

S11, S12, and S13 remained isolated across the tested thresholds. Recommendations concerning these factors should therefore be framed as legal, cultural, and informational conditions that permit the governance system to operate, rather than as empirically dominant causal levers. Their isolated positions also reflect the limited coverage of the original 33-link instrument and should not be treated as proof that they have no meaningful relationships outside the assessed network.

### **6.4 Intervention Priorities**

Priority 1—Verification responsibility and lifecycle coordination. The first priority is to assign clear responsibility for entry checks, contract-event recording, construction verification, exception classification, escalation, sanction, and credit repair. Performance dashboards should distinguish the participant that creates a risk from the organisation responsible for verifying and responding to it. This distinction is particularly important for S6, which had the highest prominence and the largest positive net relation. Verification records should identify the underlying evidence, the responsible organisation, the date of verification, the classification decision, and the action taken. These requirements turn project information into an auditable basis for operational decisions.

Priority 2—Data interoperability and smart supervision. Project owners, contractors, supervision organisations, and regulators should establish shared identifiers, minimum data fields, verification timestamps, and accountable interfaces across their information systems. Technology selection should follow the requirements of the evidence and decision chain rather than determine them. The dynamic model supported investment in a composite smart supervision capability under its stated equations, but it did not separately quantify the effects of blockchain, building information modelling, the Internet of Things, distributed sensing, or big-data analytics. S9 should also be incorporated into this information architecture through measurable environmental and resource-compliance records, although its intervention effect was not examined in the dynamic model.

Priority 3—Credit guarantees, default control, and procedural repair. Project-entry arrangements should establish proportionate performance-guarantee requirements and define the circumstances under which a guarantee may be activated. Default decisions should use explicit evidence thresholds and connect verified non-performance with notice, correction, sanction, appeal, and credit-repair procedures. This priority addresses S1 and S10 by linking assurance mechanisms to workable credit-management routines. It also prevents guarantee requirements from becoming detached from actual project events or being applied automatically to ordinary schedule changes and contractual variations.

These priorities combine structural ranking with lifecycle feasibility. They should not be interpreted as a universal sequence for every engineering project. Regulators and project organisations may need to adjust their order according to project type, institutional setting, available data infrastructure, and the severity of the risks being managed. Their purpose is to translate the analytical results into a transparent set of operational and strategic choices.



## 6.5 Limitations and Future Research

The evidence is subject to six main limitations. First, the sample consisted of 86 anonymous online submissions. The archived dataset did not contain information on occupation, professional experience, organisational affiliation, project type, engineering credit-supervision experience, or geographic location. Respondent-group and regional comparisons could therefore not be conducted. The factor-development process drew partly on China-specific engineering and regulatory policy documents. This institutional context may limit the direct transfer of the resulting factor structure to regulatory environments that were not examined. The absence of professional-profile information also prevents retrospective verification of the respondents' domain expertise.

Second, the questionnaire assessed only 33 directional relationships and used a forced-choice scale from 1 to 4 without a zero or no-influence option. Treating the unassessed relationships as structural non-links preserved the traceability of the matrix, but it made the results conditional on the coverage of the original instrument. The zero entries cannot be interpreted as respondent judgements that no substantive relationship existed.

Third, the bootstrap analysis measured score-sampling variation within the collected responses. It did not capture differences among occupational or regional groups, and it could not assess uncertainty concerning relationships that were not included in the questionnaire. The narrow bootstrap rank intervals therefore support stability within the observed 33-link topology rather than the completeness of the network.

Fourth, the ISM levels and MICMAC classifications depended on threshold selection. The local sensitivity tests retained the main deep pathway near the selected MMDE threshold but showed increasing network fragmentation at MMDE +10%. The absolute thresholds of 0.38–0.42 were too high relative to the sparse total-relation matrix to provide locally informative cutoffs.

Fifth, the system dynamics model represented S1–S7 and S11 either directly, through composite variables, or as an exogenous model input, whereas S8, S9, S10, S12, and S13 were omitted because the source model contained no corresponding equations. S6 was represented only by a composite proxy. The dynamic analysis therefore complemented the 13-factor structural model rather than validating it. In particular, the simulation could not compare direct interventions targeting S9 or S10, even though these factors occupied prominent positions in the structural results.

Sixth, several parameter values were inherited assumptions from the archived source model, and the system dynamics model had no historical calibration target. The scenario differences, Monte Carlo intervals, and trajectories over times 30–50 describe behaviour under the specified equations and parameter settings; they are not predictions of future engineering-industry credit outcomes. The use of different parameter changes in the institutional-response and technology scenarios also means that their numerical results should not be interpreted as a controlled comparison of equally intensive interventions.

Future research should collect non-identifying information on respondent roles, professional experience, project types, and regions; assess a more complete directional network using a genuine no-influence option; and examine whether the factor structure varies across institutional and project settings. An expanded system dynamics model should include S8, S9, S10, S12, and S13 explicitly and should be calibrated against longitudinal indicators where suitable data are available. Further work should also define measurable intervention costs and constraints so that governance alternatives can be compared in terms of effectiveness, timing, feasibility, and resource requirements.

## 7 Conclusions

The full-precision analysis identified insufficient credit verification by supervision units, environmental and resource compliance credit risk, and lagging credit-management methods as the three most prominent factors in the questionnaire-defined network. The MMDE-based ISM structure placed S9 at the root level and S2 and S6 at the next level, while MICMAC classified S2, S6, S8, and S9 as independent factors with relatively high driving power and low dependence.

The exploratory system dynamics model examined only the institutional-response and technology-related mechanisms represented in its equations. Under the selected parameter settings, the technology intervention produced the highest credit-index value at time 20. Reducing the regulatory response delay moderated the downturn and improved selected intermediate outcomes but did not produce a clear long-horizon advantage. At time 50, the combined scenario produced a value of 33.73, only 0.31 above the technology-only value of 33.42. This small difference is a long-horizon model diagnostic and does not establish greater stability or predict performance over a real-world 50-year period.

The findings indicate that engineering project credit-risk governance should be organised as an auditable lifecycle rather than as a stand-alone technology programme. Entry arrangements should establish guarantees, participant identities, and data permissions; contractual obligations should be translated into verifiable project events; supervision organisations should be assigned explicit responsibility for verification and escalation; and confirmed defaults should be connected to proportionate correction, sanction, appeal, and credit-repair procedures. Smart supervision

technology supports this process by improving information connectivity and monitoring capacity, but its value depends on the institutional responsibilities and decision rules within which it operates.

The study provides a traceable basis for identifying factor priorities and comparing selected intervention pathways, while its conclusions remain bounded by the 86-response sample, the questionnaire-defined 33-link network, and the exploratory, partially mapped system dynamics model. The results support decision-making within these stated boundaries and should not be interpreted as a universal causal structure or as a forecast of engineering-industry credit performance.

### Author Contributions

Conceptualization, Z.H.Z.; methodology, S.B.Z. and Z.H.Z.; software, S.B.Z.; validation, Y.Q.H.; formal analysis, S.B.Z.; investigation, Y.Q.H. and W.S.Z.; data curation, Y.Q.H. and W.S.Z.; writing—original draft preparation, S.B.Z.; writing—review and editing, Z.H.Z., Y.Q.H., and W.S.Z.; visualization, S.B.Z.; supervision, Z.H.Z.; funding acquisition, Z.H.Z. All authors have read and agreed to the published version of the manuscript.

### Data Availability

The data used to support the research findings are available from the corresponding author upon request. Individual questionnaire responses are not publicly shared because they contain respondent-level information and are subject to privacy considerations.

### Conflicts of Interest

The authors declare no conflicts of interest.

### References

- [1] D. C. North, *North, Institutions, Institutional Change and Economic Performance*. Cambridge University Press, 1990.
- [2] Y. Qiu and H. Chen, “A systematic review of the knowledge domain of institutional theory in construction project management,” *Eng. Constr. Archit. Manage.*, vol. 30, no. 6, pp. 2523–2544, 2023. <https://doi.org/10.1108/ECAM-08-2021-0754>
- [3] F. K. Tetteh, K. O. Kwateng, and H. Obiri-Yeboah, “Understanding green building practices adoption in the construction industry: An extension of institutional theory,” *Property Manage.*, vol. 43, no. 5, pp. 806–831, 2025. <https://doi.org/10.1108/PM-09-2024-0096>
- [4] I. Ayres and J. Braithwaite, “Tripartism: Regulatory capture and empowerment,” *Law Social Inquiry*, vol. 16, no. 3, pp. 435–496, 1991. <https://doi.org/10.1111/j.1747-4469.1991.tb00291.x>
- [5] Z. Xiufan, Y. Shi, and L. Meng, “Research on the mechanism and path of the coupling of digital technology and environmental regulation to promote urban green efficiency,” *Sustain. Cities Soc.*, vol. 116, p. 105906, 2024. <https://doi.org/10.1016/j.scs.2024.105906>
- [6] N. Berente and S. Seidel, “Digital technologies: Carrier or trigger for institutional change in digital transformation,” in *Digital Transformation and Institutional Theory*. Emerald Publishing Limited, 2022. <https://doi.org/10.1108/S0733-558X20220000083008>
- [7] J. Zhang, M. Zhang, P. Ballesteros-Pérez, and S. P. Philbin, “A new perspective to evaluate the antecedent path of adoption of digital technologies in major projects of construction industry: A case study in China,” *Dev. Built Environ.*, vol. 14, p. 100160, 2023. <https://doi.org/10.1016/j.dibe.2023.100160>
- [8] S. Nanayakkara, S. Perera, S. Senaratne, G. T. Weerasuriya, and H. M. N. D. Bandara, “Blockchain and smart contracts: A solution for payment issues in construction supply chains,” *Informatics*, vol. 8, no. 2, p. 36, 2021. <https://doi.org/10.3390/informatics8020036>
- [9] W. Lu, L. Wu, and R. Zhao, “Rebuilding trust in the construction industry: A blockchain-based deployment framework,” *Int. J. Constr. Manage.*, vol. 23, no. 8, pp. 1405–1416, 2023. <https://doi.org/10.1080/15623599.2021.1974683>
- [10] X. Hu, G. Olgun, and R. H. Assaad, “An intelligent BIM-enabled digital twin framework for real-time structural health monitoring using wireless IoT sensing, digital signal processing, and structural analysis,” *Expert Syst. Appl.*, vol. 252, p. 124204, 2024. <https://doi.org/10.1016/j.eswa.2024.124204>
- [11] L. Sun, H. Sun, W. Zhang, and Y. Li, “Hybrid monitoring methodology: A model-data integrated digital twin framework for structural health monitoring and full-field virtual sensing,” *Adv. Eng. Inform.*, vol. 60, p. 102386, 2024. <https://doi.org/10.1016/j.aei.2024.102386>
- [12] S. D. Datta, M. Islam, M. H. R. Sobuz, S. Ahmed, and M. Kar, “Artificial intelligence and machine learning applications in the project lifecycle of the construction industry: A comprehensive review,” *Heliyon*, vol. 10, no. 5, p. e26888, 2024. <https://doi.org/10.1016/j.heliyon.2024.e26888>

- [13] S. O. Abioye, L. O. Oyedele, L. Akanbi, A. Ajayi, J. M. D. Delgado, M. Bilal, O. O. Akinade, and A. Ahmed, "Artificial intelligence in the construction industry: A review of present status, opportunities and future challenges," *J. Build. Eng.*, vol. 44, p. 103299, 2021. <https://doi.org/10.1016/j.job.2021.103299>
- [14] F. Antoniou and A. V. Tsioulpa, "Assessing the delay, cost, and quality risks of claims on construction contract performance," *Buildings*, vol. 14, no. 2, p. 333, 2024. <https://doi.org/10.3390/buildings14020333>
- [15] Y. Jiang, S. Su, S. Zhao, R. Y. Zhong, W. Qiu, M. J. Skibniewski, I. Brilakis, and G. Q. Huang, "Digital twin-enabled synchronized construction management: A roadmap from Construction 4.0 towards future prospect," *Dev. Built Environ.*, vol. 19, p. 100512, 2024. <https://doi.org/10.1016/j.dibe.2024.100512>
- [16] L. G. Tornatzky, M. Fleischer, and A. K. Chakrabarti, *The Processes of Technological Innovation*. Lexington Books, 1990.
- [17] H. Luo, M. Das, J. Wang, and J. C. P. Cheng, "Construction payment automation through smart contract-based blockchain framework," in *2019 Proceedings of the 36th ISARC*, Banff, Canada, 2019, pp. 1254–1260. <https://doi.org/10.22260/ISARC2019/0168>
- [18] A. Gómez-Cabrera, L. Gutierrez-Bucheli, and S. Muñoz, "Causes of time and cost overruns in construction projects: A scoping review," *Int. J. Constr. Manage.*, vol. 24, no. 10, pp. 1107–1125, 2024. <https://doi.org/10.1080/15623599.2023.2252288>
- [19] A. Gondia, A. Moussa, M. Ezzeldin, and W. El-Dakhakhni, "Machine learning-based construction site dynamic risk models," *Technol. Forecast. Social Change*, vol. 189, p. 122347, 2023. <https://doi.org/10.1016/j.techfore.2023.122347>
- [20] C. Liu, L. Li, Y. Qiang, and S. Zhang, "Predicting construction accidents on sites: An improved atomic search optimization algorithm approach," *Eng. Rep.*, vol. 6, no. 5, p. e12773, 2024. <https://doi.org/10.1002/eng2.12773>
- [21] H. Wu, P. Zhang, H. Li, B. Zhong, I. W. H. Fung, and Y. Y. R. Lee, "Blockchain technology in the construction industry: Current status, challenges, and future directions," *J. Constr. Eng. Manage.*, vol. 148, no. 10, p. 03122007, 2022. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0002380](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002380)
- [22] Z. Liu and N. Wang, "The effects of emerging digital technologies on construction project resilience: The mediating role of relational governance," *Build. Res. Inf.*, vol. 54, no. 1, pp. 118–134, 2026. <https://doi.org/10.1080/09613218.2025.2482961>
- [23] A. M. Alsugair, K. S. Al-Gahtani, N. M. Alsanabani, G. M. Hommadi, and M. I. Alawshan, "An integrated DEMATEL and system dynamic model for project cost prediction," *Heliyon*, vol. 10, no. 4, p. e26166, 2024. <https://doi.org/10.1016/j.heliyon.2024.e26166>
- [24] B. O. Ajayi and T. Chinda, "Impact of construction delay-controlling parameters on project schedule: DEMATEL-system dynamics modeling approach," *Front. Built Environ.*, vol. 8, p. 799314, 2022. <https://doi.org/10.3389/fbuil.2022.799314>
- [25] J. Cheng, X. Fan, L. Tian, and J. Li, "Uncovering the drivers and pathways of carbon emissions in smart city: An integrated DEMATEL–ISM–system dynamics approach," *Buildings*, vol. 16, no. 1, p. 99, 2026. <https://doi.org/10.3390/buildings16010099>
- [26] M. Rostamnezhad, F. Nasirzadeh, M. Khanzadi, M. J. Jarban, and M. Ghayoumian, "Modeling social sustainability in construction projects by integrating system dynamics and fuzzy-DEMATEL method: A case study of highway project," *Eng. Constr. Archit. Manage.*, vol. 27, no. 7, pp. 1595–1618, 2020. <https://doi.org/10.1108/ECAM-01-2018-0031>
- [27] X. Pan, Y. Wei, S. K. Mangla, and M. Song, "How do ESG practices promote collaboration on green investment for supply chain enterprises in China?" *Transp. Res. Part E: Logist. Transp. Rev.*, vol. 198, p. 104118, 2025. <https://doi.org/10.1016/j.tre.2025.104118>
- [28] T. Patel, H. Bapat, and D. Patel, "Assessing barriers of automation and robotics adoption in the Indian construction industry: A fuzzy DEMATEL approach," *Constr. Innov.*, vol. 26, no. 3, pp. 935–957, 2026. <https://doi.org/10.1108/CI-04-2024-0120>
- [29] W. Zhao, K. Wang, F. Guo, J. L. Hao, Q. Zhang, and J. Qi, "Unraveling barriers to digital twin adoption under Construction 4.0: A DEMATEL-ISM-MICMAC approach," *J. Build. Design Environ.*, vol. 3, p. 202438, 2025. <https://doi.org/10.70401/jbde.2025.0002>
- [30] W. Lu, L. Wu, R. Zhao, X. Li, and F. Xue, "Blockchain technology for governmental supervision of construction work: Learning from digital currency electronic payment systems," *J. Constr. Eng. Manage.*, vol. 147, no. 10, 2021. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0002148](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002148)
- [31] T. D. Akinosho, L. O. Oyedele, M. Bilal, A. O. Ajayi, M. D. Delgado, O. O. Akinade, and A. A. Ahmed, "Deep learning in the construction industry: A review of present status and future innovations," *J. Build. Eng.*, vol. 32, p. 101827, 2020. <https://doi.org/10.1016/j.job.2020.101827>
- [32] M. Kamari and Y. Ham, "AI-based risk assessment for construction site disaster preparedness through deep learning-based digital twinning," *Autom. Constr.*, vol. 134, p. 104091, 2022. <https://doi.org/10.1016/j.autcon.2022.104091>

- [33] Z. Zhou, L. Wei, J. Yuan, J. Cui, Z. Zhang, W. Zhuo, and D. Lin, "Construction safety management in the data-rich era: A hybrid review based upon three perspectives of nature of dataset, machine learning approach, and research topic," *Adv. Eng. Inform.*, vol. 58, p. 102144, 2023. <https://doi.org/10.1016/j.aei.2023.102144>
- [34] S. Wan, Y. Liu, G. Ding, G. Runeson, and M. Er, "Risk allocation for energy performance contract from the perspective of incomplete contract: A study of commercial buildings in China," *Int. J. Climate Change Strateg. Manage.*, vol. 15, no. 4, pp. 457–478, 2023. <https://doi.org/10.1108/IJCCSM-11-2021-0130>
- [35] T. T. Okanlawon, L. O. Oyewobi, and R. A. Jimoh, "Effect of blockchain technology adoption on construction supply chain: A structural equation modelling (SEM) approach," *J. Facil. Manage.*, vol. 23, no. 3, pp. 407–428, 2025. <https://doi.org/10.1108/JFM-07-2023-0077>
- [36] A. K. Singh, V. R. P. Kumar, G. Dehdasht, S. R. Mohandes, P. Manu, and F. P. Rahimian, "Investigating barriers to blockchain adoption in construction supply chain management: A fuzzy-based MCDM approach," *Technol. Forecast. Social Change*, vol. 196, p. 122849, 2023. <https://doi.org/10.1016/j.techfore.2023.122849>
- [37] A. K. Singh, V. R. P. Kumar, M. Irfan, S. R. Mohandes, and U. Awan, "Revealing the barriers of blockchain technology for supply chain transparency and sustainability in the construction industry: An application of Pythagorean FAHP methods," *Sustainability*, vol. 15, no. 13, p. 10681, 2023. <https://doi.org/10.3390/su151310681>
- [38] S. Ahmadiheykhsarmast, S. G. Senji, and R. Sonmez, "Decentralized tendering of construction projects using blockchain-based smart contracts and storage systems," *Autom. Constr.*, vol. 151, p. 104900, 2023. <https://doi.org/10.1016/j.autcon.2023.104900>
- [39] Z. Liu, L. Zhou, Q. K. Qian, H. Visscher, and G. Zhang, "Towards sustainable development: Analyzing policy frameworks and barriers in China's green construction transition," *Sustain. Energy Technol. Assess.*, vol. 82, p. 104499, 2025. <https://doi.org/10.1016/j.seta.2025.104499>
- [40] C. Zhang, K. Hu, and X. Liu, "Digital transformation of construction industry under government reward and punishment mechanism based on evolutionary game modeling," *J. Asian Archit. Build. Eng.*, vol. 24, no. 5, pp. 3845–3862, 2025. <https://doi.org/10.1080/13467581.2024.2396594>
- [41] J. Franke and P. Gailhofer, "Data governance and regulation for sustainable smart cities," *Front. Sustain. Cities*, vol. 3, p. 763788, 2021. <https://doi.org/10.3389/frsc.2021.763788>
- [42] S. Sorooshian, S. M. Jamali, and N. Ale Ebrahim, "Performance of the decision-making trial and evaluation laboratory," *AIMS Math.*, vol. 8, no. 3, pp. 7490–7514, 2023. <https://doi.org/10.3934/math.2023376>
- [43] C. W. Li and G. H. Tzeng, "Identification of a threshold value for the DEMATEL method using the maximum mean de-entropy algorithm to find critical services provided by a semiconductor intellectual property mall," *Expert Syst. Appl.*, vol. 36, no. 6, pp. 9891–9898, 2009. <https://doi.org/10.1016/j.eswa.2009.01.073>
- [44] K. J. Arrow, "The economic implications of learning by doing," *Rev. Econ. Stud.*, vol. 29, no. 3, pp. 155–173, 1962. <https://doi.org/10.2307/2295952>
- [45] E. M. Rogers, *Diffusion of Innovations*, 5th ed. New York, NY, USA: Free Press, 2003.
- [46] J. D. Sterman, *Business Dynamics: Systems Thinking and Modeling for a Complex World*. McGraw Hill Higher Education, 2000.